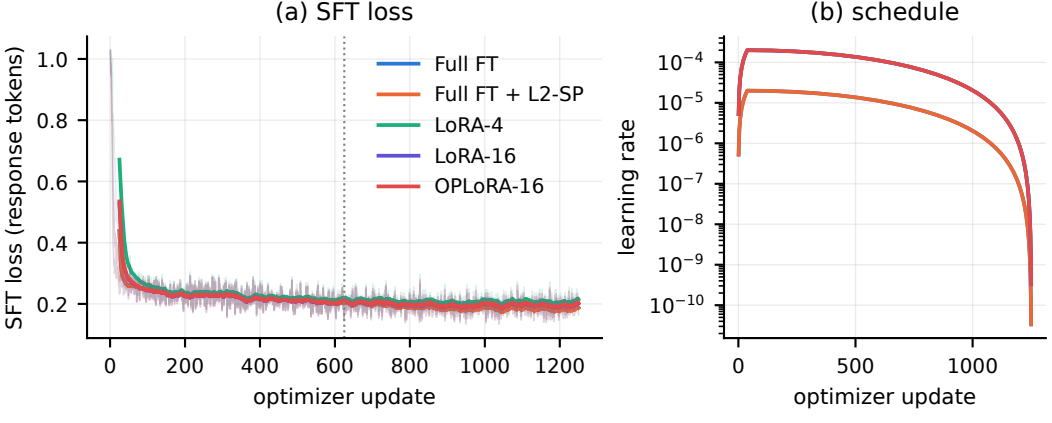

(a) SFT loss
SFT loss (response tokens)
1.0
0.8
0.6
0.4
0.2
0
200
400
600
800
1000
1200
optimizer update
Full FT
Full FT + L2-SP
LoRA-4
LoRA-16
OPLoRA-16
(b) schedule
learning rate
10^{-4}
10^{-5}
10^{-6}
10^{-7}
10^{-8}
10^{-9}
10^{-10}
0
500
1000
optimizer update